

HW2

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Q1

Suppose that $\underline{y}$ is an $n \times 1$ observable random vector that follows Model II.

(a)

If $\underline{\lambda}'_1 \underline{\beta}$ and $\underline{\lambda}'_2 \underline{\beta}$ are both estimable, is $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta}$ necessarily estimable? Explain your answer.

Answer.

Yes. If $\underline{\lambda}'_1 \underline{\beta}$ and $\underline{\lambda}'_2 \underline{\beta}$ are both estimable, then there exist vectors $\underline{a}_1$ and $\underline{a}_2$ such that $E(\underline{a}_1 \underline{y}) = \underline{\lambda}'_1 \underline{\beta}$ and $E(\underline{a}_2 \underline{y}) = \underline{\lambda}'_2 \underline{\beta}$. It would then follow that $E(\underline{a}_1 \underline{y} + \underline{a}_2 \underline{y}) = \underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta}$.

(b)

If $\underline{\lambda}'_1 \underline{\beta}$ and $\underline{\lambda}'_2 \underline{\beta}$ are both nonestimable, is $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta}$ necessarily nonestimable? Explain your answer.

Answer.

No. Suppose that $\underline{\lambda}'_1 \underline{\beta}$ is nonestimable. Then, $-\underline{\lambda}'_1 \underline{\beta}$ is also nonestimable. However, $\underline{\lambda}'_1 \underline{\beta} + (-\underline{\lambda}'_1 \underline{\beta}) = 0$, and this quantity is trivially estimable.

(c)

Can $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta}$ be estimable if $\underline{\lambda}'_1 \underline{\beta}$ is estimable and $\underline{\lambda}'_2 \underline{\beta}$ is nonestimable? Explain your answer.

Answer.

No. Suppose $\underline{\lambda}'_1 \underline{\beta}$ and $\underline{\lambda}'_2 \underline{\beta}$ are both estimable. Then, necessarily, $\underline{\lambda}'_1 \underline{\beta} = \underline{\lambda}'_1 \underline{\beta} + (\underline{\lambda}'_2 \underline{\beta} - \underline{\lambda}'_2 \underline{\beta}) = (\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta}) - \underline{\lambda}'_2 \underline{\beta}$ must also be estimable.

Q2

Take the model to be Model VII and suppose that

$$\underline{y} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \quad \text{and} \quad X = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

(a)

Find the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and on $\mathcal{C}^\perp(X)$.

Answer.

(b)

Determine $\text{rank}(X'X)$.

Answer.

Via Lemma 2.3, $\text{rank}(X'X) = \text{rank}(X) = 2$.

(c)

Use the following method and find a generalized inverse of $X'X$.

Let A be an $n \times m$ matrix of rank $r \geq 1$. Let B be a nonsingular $r \times r$ submatrix of A . For matrix A , replace the entries of B by those of $(B^{-1})'$, and replace all other entries of A by 0. Then the transpose of the resulting matrix is a generalized inverse of A .

Answer.

(d)

Compute a solution $\hat{\underline{\beta}}$ to the normal equations.

Answer.

(e)

Check that the vectors $X\hat{\underline{\beta}}$ and $\underline{y} - X\hat{\underline{\beta}}$ agree with your results in part (a).

Answer.

Q3

For Model VII, partition $\underline{y}$, X , and $\underline{\epsilon}$ correspondingly as follows:

$$\underline{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \quad \text{and} \quad \underline{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \end{pmatrix}.$$

Suppose that the row spaces of X_1 and X_2 are essentially disjoint, that is, $\mathcal{R}(X_1) \cap \mathcal{R}(X_2) = \{0'\}$. Show that, if $\underline{\lambda}' \in \mathcal{R}(X_1)$, then the least squares estimate of $\underline{\lambda}'\underline{\beta}$ based on the model $\underline{y}_1 = X_1\underline{\beta} + \underline{\epsilon}_1$ is the same as the least squares estimate of $\underline{\lambda}'\underline{\beta}$ based on the model $\underline{y} = X\underline{\beta} + \underline{\epsilon}$.

Hint: Under the full model, the LSE is $\underline{a}'X'\underline{y} = \underline{a}'X'_1\underline{y}_1 + \underline{a}'X'_2\underline{y}_2$ and, under the partial model, the LSE is $\underline{b}'X'_1\underline{y}_1$ where here $X'_1X_1\underline{a} + X'_2X_2\underline{a} = X'X\underline{a} = \underline{\lambda} = X'_1X_1\underline{b}$ (why?); in the last equation, move terms involving X'_2 and X'_1 to different sides.

Answer.

Q4

Show that, under Model II, $\underline{a}'\underline{y}$ is a BLUE of its expectation if and only if $\underline{a} \in \mathcal{C}(X)$.

Answer.

Q5

(a)

Show that, under Model II, the problem of determining the BLUE $\underline{a}'\underline{y}$ of an estimable function $\underline{\lambda}'\underline{\beta}$ is equivalent to that of minimizing $\underline{a}'\underline{a}$ subject to the constraints $\underline{a}'X = \underline{\lambda}'$.

Answer.

(b)

The Lagrangian function associated with this minimization problem is

$$L(\underline{a}, \underline{\rho}) = \underline{a}'\underline{a} + 2\underline{\rho}'(X'\underline{a} - \underline{\lambda}).$$

Use the vector calculus to differentiate the Lagrangian function with respect to $\underline{a}$ and the Lagrangian multipliers, and set these derivatives equal to zero.

Answer.

(c)

Show that the resulting equations have a solution if and only if the conjugate normal equations are consistent.

Answer.

(d)

Solve your equations for the minimizing value of $\underline{a}$.

Answer.